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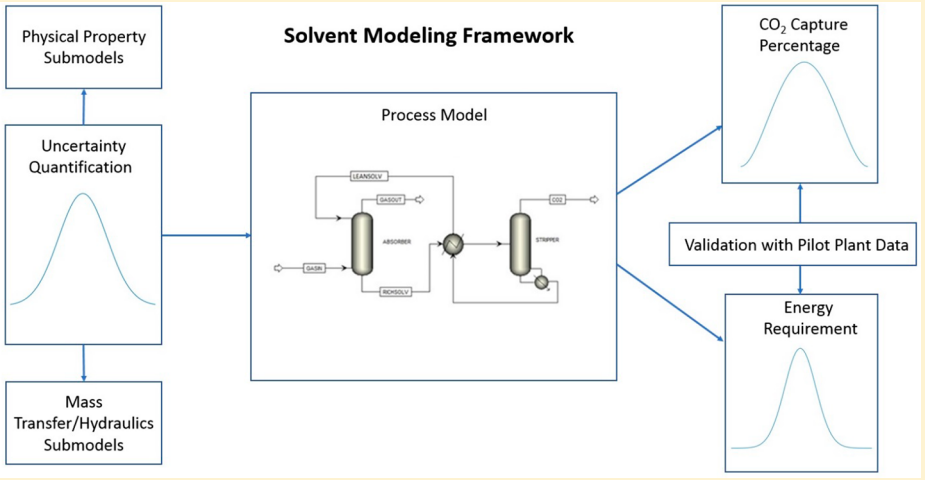
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Development of a Rigorous Modeling Framework for Solvent-Based CO_2 Capture. Part 2: Steady-State Validation and Uncertainty Quantification with Pilot Plant Data

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Supporting Information



ABSTRACT: The U.S. DOE's Carbon Capture Simulation Initiative (CCSI) has a strong focus on the development of state of the art process models for accelerating the development and commercialization of postcombustion carbon capture system technologies. One of CCSI's goals is the development of a process model that will serve not only as a definitive reference for benchmarking of the performance of solvent-based CO_2 capture systems but also as a framework for the development of highly predictive models of advanced solvent systems. In Part 1 of this paper and previous work, submodels for the system were developed, including those for physical properties, kinetics, mass transfer, and column hydraulics, by calibrating model parameters to fit relevant experimental data. For individual submodels, a Bayesian inference methodology was used to refine the estimates of the parameter values and to quantify the parametric uncertainty of the models. This work is focused on incorporating these submodels into a complete process model and validating this model with large scale pilot plant data from the Pilot Solvent Test Unit (PSTU) at the National Carbon Capture Center (NCCC). The model has been validated with data representing a wide range of operating conditions for absorber and stripper columns, including variable packing height and presence of intercooling in the absorber. The uncertainty in the solvent composition is measured by comparing the process measurements at NCCC to standard laboratory techniques of a known uncertainty. Through a sensitivity study, this measurement uncertainty is used to provide insight into some discrepancy between model and data. Parametric uncertainty for various submodel parameters has been propagated through the process model to assess the resulting uncertainty in the key model outputs. Finally, a variance-based sensitivity analysis is used to provide insight into the relative contributions of parameters from various submodels to the overall uncertainty of the process outputs in various operating regimes.

1. INTRODUCTION

The U.S. Department of Energy's Carbon Capture Simulation Initiative (CCSI) focused on developing computational tools and models to accelerate the development and commercialization of CO_2 capture technologies.[1] One key goal of CCSI was

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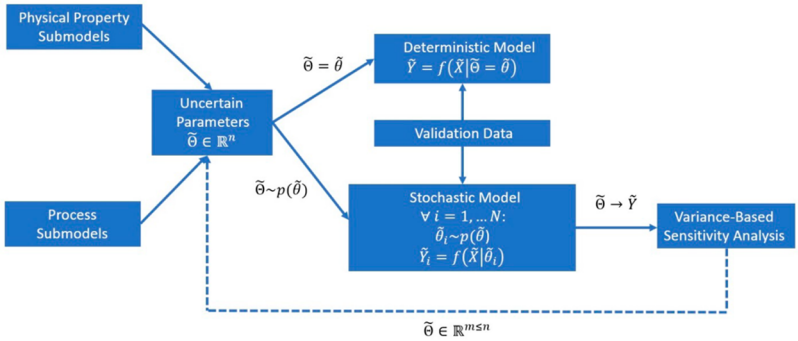


Figure 1. Schematic of process model development approach.

to develop a modeling approach for solvent-based CO_2 capture that can be readily applied to any solvent system. Specifically, it is desired to use this framework for accelerating the model development and scale-up of novel solvent systems. However, the baseline solvent system used for the development of the solvent modeling framework is aqueous monoethanolamine (MEA), due to the relatively large amount of property and process data available in the open literature for this system. Aqueous solutions with 30 wt % MEA have been the industrial standard for acid gas removal from solvents since 1970.[2] Our previous work,[3,4] in addition to Part 1 of this two-part paper, have focused on development and uncertainty quantification (UQ) of the submodels that comprise the overall process model of CO_2 capture by aqueous MEA. In Part 1, development of rigorous submodels for mass transfer and hydraulics and their UQ is described. In Part 2, the focus is on model validation using steady-state data obtained from the Pilot Solvent Test Unit (PSTU) at the National Carbon Capture Center (NCCC) in Wilsonville, AL. This pilot plant, with a CO_2 capture capacity of approximately 10 ton/day and electric output 0.5 MWe, is relatively large in comparison to most of the solvent-based CO_2 capture systems for which data is available in the open literature. As part of the model validation process, uncertainties in the integrated model are quantified.

Deterministic process models of the CO_2 capture system with MEA are available in the open literature, including comparison with pilot plant data. Most of the existing validation work, however, is based on pilot plants of a smaller scale or over narrower operating ranges. In the work of Luo et al.,[5] models from leading commercial simulators are compared with data from four separate pilot plants, with significant discrepancy in the results generated by the various models for both the absorber and stripper. Previous MEA solvent test campaigns have also been performed and used for model validation at pilot plant (0.1 MWe) at the J.J. Pickle Research Center of The University of Texas at Austin.[6−8] The pilot plant (0.2 MWe) at SaskPower's Boundary Dam Power Station has also been the subject of previous validation work with MEA.[9,10] Data from the pilot plant (~0.01 MWe) at the University of Kaiserslautern have been reported for various solvent systems, with aqueous MEA generally considered as a baseline.[11−17] Multiple papers from Moser et al.[18−20] have presented information regarding a long-term MEA test at a power plant (~0.35 MWe) at RWE Power's power station in Niederaussem, Germany. In addition to those mentioned here, various other MEA model validation studies with small-scale pilot plant data have been reported.[21−30] Some work on MEA model validation with large-scale pilot plant data (~1.2 MWe) is available for the CASTOR project pilot plant at the Esbjerg power station in Denmark, although validation over a large range of operating conditions for both columns is not shown.[31,32] In summary, validation with the data from large pilot plants such as NCCC is very scarce. Furthermore, models have typically been validated over a narrow range of operating conditions. However, when the host power plant must ramp up or down, there can be significant variations in the operating conditions of the CO_2 capture plant. Furthermore, variable CO_2 capture, as opposed to the standard 90% capture condition, can be exploited as a strategy for load-following. Therefore, it is desired that the model be validated with the pilot plant data over a wide range of operating conditions including the CO_2 capture percentage.

Existing models of the MEA-based CO_2 capture process are deterministic in nature, meaning that rigorous parametric uncertainty quantification for the system has not been considered; however, some work has studied the sensitivity of the overall process model to process variables and parameters in the underlying submodels. Nuchitprasittichai and Cremaschi[33] performed a sensitivity analysis in which the effect of fluctuations in flue gas CO_2 concentration and utility costs on process economics is studied for various amine-based CO_2 capture systems. Tönnies et al.[34] performed a sensitivity study for the MEA system with respect to input parameters describing fluid dynamics and physical and chemical properties of the system. In the MEA system absorber model validation work of Tobiesen et al.,[24] a parametric sensitivity study concluded that the match between experimental data and model predications are particularly sensitive to the equilibrium model. deMontigny et al.[35] studied the effects of operating variables such as liquid and gas flow, solvent CO_2 loading, MEA concentration, and CO_2 partial pressure on the overall mass transfer coefficient. An optimization study was performed by Dinca and Badea[36] in which process variables such as solvent flow rate and composition and operating temperatures were manipulated to minimize heat required for MEA regeneration and maximize CO_2 capture efficiency. Another optimization study focused on maximizing the ratio between absorbed CO_2 and total heating and cooling utility and the ratio between absorbed CO_2 and amine flow rate by manipulating the key process variables.[37]

The contributions of this work may be summarized as follows. (1) A modeling framework for modeling and uncertainty quantification of solvent-based CO_2 capture systems has been outlined and applied to a baseline aqueous MEA system. (2) A model of a large-scale pilot plant (PSTU at

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NCCC) has been developed in Aspen Plus, consisting of physical property, kinetic, mass transfer, and hydraulic submodels reported in our previous work. (3) This model has been validated with experimental data that span a wide range of operating conditions, including liquid and gas flow rates and compositions, stripper reboiler duty, and packing height in the absorber column. (4) Probability density functions (PDFs) describing parametric uncertainty from the aforementioned submodels are propagated through the process model for various test cases, allowing estimation of the corresponding uncertainty in key process variables. The effect of uncertainty of individual parameters on process output variable uncertainty is assessed for various operating conditions. (5) Uncertainty in the measurement of the solvent composition has been experimentally determined, and the effect of the measurement uncertainty on the process output variables is assessed.

2. METHODOLOGY

2.1. Modeling Framework. Although this work is primarily focused on model validation with pilot plant data for the MEA system specifically, it is important to note that the model has been developed with a systematic framework that may be applied to a generic solvent system. The overall process is shown schematically in Figure 1.

The overall process model is composed of various submodels, including physical property (e.g., thermodynamic and transport properties for the solvent system) and process (e.g., kinetic, mass transfer, hydraulics) models. The development of these models includes a method for identifying uncertain parameters through parameter selection and uncertainty quantification with Bayesian inference; this results in a combined set of uncertain parameters ($\tilde{\Theta}$) that incorporates all submodels. Both deterministic and stochastic versions of the overall process model are developed and validated with process data, which may be from either bench or pilot scale. The major difference between the deterministic and stochastic models is related to the treatment of the uncertain parameters. For the deterministic model, the parameters are treated as a set of point values ($\tilde{\theta}$), generally the maximum likelihood estimators that are obtained from regression of the submodel forms with relevant experimental data. In the stochastic model, these parameters are represented as probability density functions (PDFs) that are estimated through a Bayesian framework. When evaluating the stochastic model for a fixed set of input variables $\tilde{X}$, a sample $\tilde{\theta}_i$ is drawn from the joint parameter distribution and used to generate a set of model output values ($\tilde{Y}_i$) for that specific instantiation from the distribution. The overall process model is validated through comparison with process data (e.g., operation data for the absorber and stripper columns at a pilot plant) both deterministically and stochastically. The common validation approach is to compare plant data with deterministic model output over a range of operating conditions (variable $\tilde{X}$) and check for accuracy in prediction. This methodology expands upon this approach by using the stochastic model to predict ranges of output variable values in order to allow for model uncertainty to be taken into consideration during the validation process. It is important to note that all parameter values and distributions are characterized independently of the validation data, meaning that no parameters are tuned to improve the fit of the process model to the data. Although the modeling approach could be modified from the version shown in Figure 1 to allow for use of process data to update submodel parameter values and distributions, this will be left to future work.

In the variance-based sensitivity analysis, the mapping (denoted as $\tilde{\Theta} \rightarrow \tilde{Y}$) of the output variable values to the uncertain parameter values for a specific set of input values is used to decompose the variance in the model output into individual contributions of parameters and their interactions. One such analysis is described in the work of Sobol,[38] and will be applied to this work. For each output variable ($Y \in \tilde{Y}$) and each parameter ($\Theta \in \tilde{\Theta}$) or combination of parameters, an index value is calculated; larger index values indicate that the output variable has relatively higher sensitivity to that particular combination of parameters. For simplifying this analysis, the total sensitivity index for a parameter can be interpreted as the index for that individual parameter and the sum of its interactions with all other groups of parameters. Parameters with very low indices for all outputs over a wide range of input conditions can be assumed to have little effect on the uncertainty in the model output. Therefore, the uncertainty of these parameters may be neglected and the parameters treated as point values, when evaluating the stochastic model in future applications. The use of the results from the variance-based sensitivity analysis to downselect the set of uncertain parameters is an optional step in the process, as denoted by the dashed line in Figure 1. Due to the computational expense of this procedure, particularly when considering multiple outputs over a wide range of input variables, the sensitivity analysis may be performed with a surrogate model replacing the actual simulation model $f(\tilde{X}|\tilde{\Theta})$.

This framework has been applied to an aqueous MEA system as a baseline, with physical property submodels described in previous work[3,4] and other process submodels in Part 1 of this work. The subsequent sections of this paper will focus on the validation of the deterministic and stochastic process models with pilot plant data along with the variance-based sensitivity analysis, thus completing the demonstration of this methodology for the baseline MEA solvent.

2.2. NCCC Steady State Test. The validation data used in this work were obtained from a test campaign at the PSTU at NCCC in the summer of 2014. Although this work is focused on validation at steady-state, dynamic data were also collected during this testing period. A number of important inputs, disturbances, and operating conditions were selected for the steady-state test cases, which include the following variables: (1) solvent flow rate, (2) flue gas flow rate, (3) flue gas composition, (4) lean solvent loading, (5) number of absorber beds, and (6) presence/absence of intercooler.

The PSTU at NCCC has a CO_2 capacity of 10 ton/day and a maximum reboiler duty of about 700 kW. The absorber and stripper columns consist of three and two beds, respectively, with each bed containing approximately 6 m of MellapakPlus 252Y structured packing. The column diameter is approximately 0.64 m for the absorber and 0.59 m for the stripper. Multiple solvent inlets are present in the absorber column so that the number of beds may be varied between 1 and 3, and intercoolers are present between the absorber beds.

A set of 31 test runs was proposed using a test matrix based on a statistical experimental design in which variables are varied simultaneously to cover a wide range of operating conditions to obtain a final data set that is suitable for rigorous model validation. Figure 2 shows an example of the test matrix

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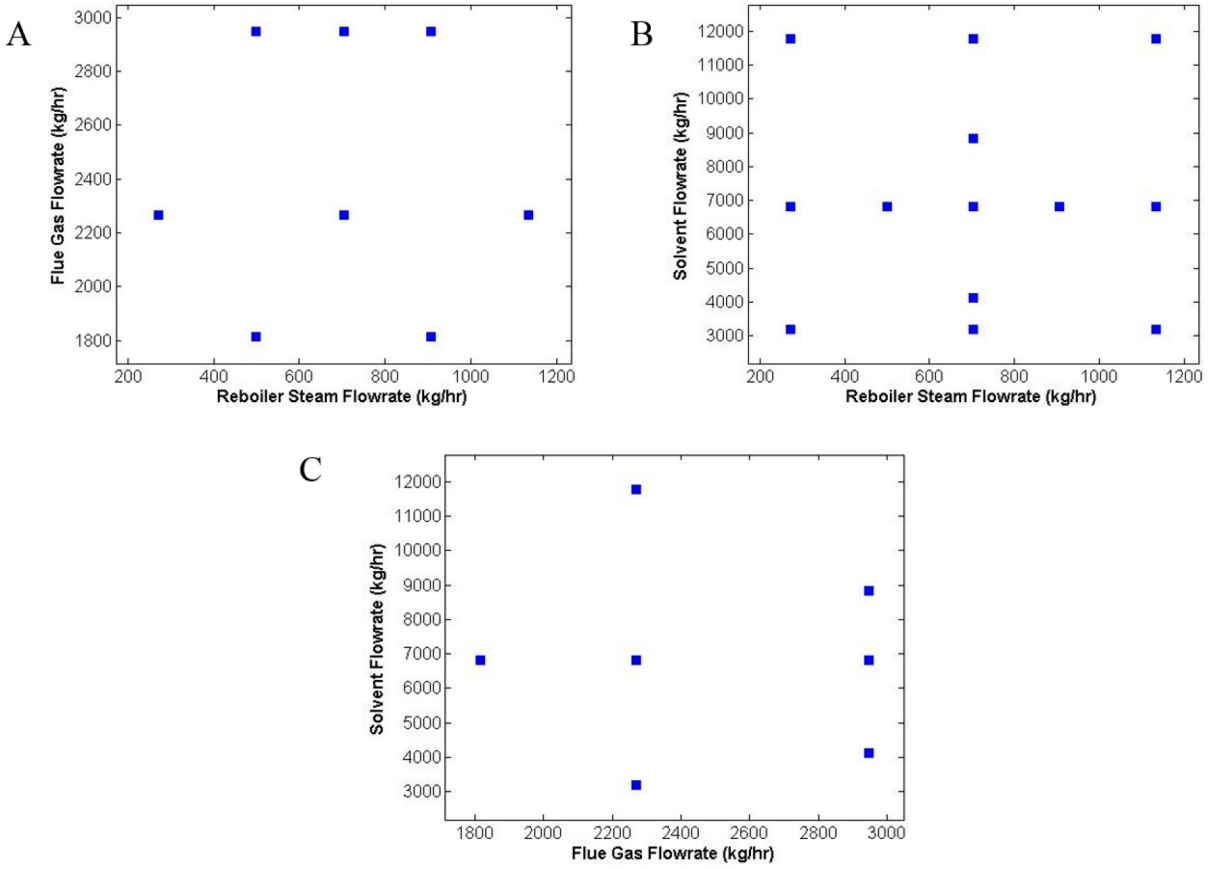


Figure 2. Simultaneous variation of flow rates in proposed test matrix for (A) flue gas flow rate vs reboiler steam flow rate, (B) solvent flow rate vs reboiler steam flow rate, and (C) solvent flow rate vs flue gas flow rate.

Table 1. Summary of NCCC Steady State Data

case no.	lean solvent flow rate (kg/h)	gas flow rate (kg/h)	lean solvent CO_2 Loading (mol CO_2/MEA)	nominal lean solvent MEA weight fraction	CO_2 mol %	number of beds (intercoolers)	reboiler duty (kW)	CO_2 capture percentage
K1	6804	2248	0.145	0.298	11.55	3 (2)	431	99.91
K2	11794	2243	0.247	0.312	11.40	3 (2)	430	99.49
K3	3175	2242	0.091	0.295	10.58	3 (2)	427	83.57
K4	3175	2243	0.083	0.310	11.45	3 (2)	427	78.10
K5	6804	2235	0.108	0.306	9.19	3 (2)	673	99.53
K6	6804	2237	0.347	0.307	9.12	3 (2)	173	59.03
K7	11791	2233	0.399	0.288	9.18	3 (2)	170	54.76
K8	11643	2237	0.154	0.285	9.20	3 (2)	677	98.07
K9	3175	2232	0.239	0.311	9.22	3 (2)	166	55.48
K10	3175	2237	0.062	0.297	9.18	3 (2)	671	98.43
K11	6804	2237	0.161	0.293	10.01	3 (2)	425	99.75
K12	6804	2231	0.160	0.293	8.24	3 (2)	422	99.61
K13	6804	2238	0.164	0.303	9.35	3 (0)	419	97.98
K14	8845	2895	0.224	0.303	9.01	3 (2)	420	98.27
K15	8845	2908	0.224	0.313	9.07	3 (2)	420	99.42
K16	4128	2903	0.124	0.329	9.21	3 (2)	419	93.54
K17	6804	2240	0.168	0.307	9.19	2 (0)	418	97.61
K18	6804	2271	0.141	0.302	10.19	1 (0)	425	92.85
K19	11793	1440	0.184	0.278	11.00	1 (0)	427	98.21
K20	3175	1324	0.075	0.276	10.98	1 (0)	438	95.55
K21	3175	1366	0.074	0.271	10.18	2 (0)	437	96.32
K22	6804	1379	0.130	0.281	10.92	2 (1)	427	99.49
K23	6804	1418	0.138	0.281	10.13	2 (1)	427	99.58

approach in terms of simultaneous variation of flow rates of solvent, flue gas, and steam to the reboiler in the stripper.

A total of 23 steady-state runs were completed from the NCCC steady state test. The first 17 test runs were conducted

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between 06/02/2014 and 06/12/2014. The runs were suspended due to process equipment failure and were resumed on 08/20/2014, starting with dynamic test runs followed by six additional steady-state runs. The tests were concluded on 08/25/2014. The measured values of several key operating variables are shown for each case in Table 1. More details on each case are given in the Supporting Information.

As shown in the table above, the apparent composition of the MEA-CO_2-H_2O solvent system is represented by two independent variables, the nominal MEA weight fraction and CO_2 loading. Nominal MEA concentration is considered the amount of amine solvent that would be present if no CO_2 was present, expressed on a mass basis. CO_2 loading is a ratio of the molar concentration of CO_2 to MEA. These quantities are calculated in terms of the apparent species weight fractions (w_i) and molecular weights (MW_i) as

$$w^*_{MEA} = \frac{w_{MEA}}{1 - w_{CO_2}} \quad (1)$$

$$\alpha = \frac{w_{CO_2}}{w_{MEA}} \times \frac{MW_{MEA}}{MW_{CO_2}} \quad (2)$$

2.3. Process Model Development. A process model for the aqueous MEA system has been developed in Aspen Plus, incorporating submodels from Part 1 of this work as well as the previous work on the viscosity, molar volume, and surface tension models[3] and the thermodynamic framework.[4] The thermodynamic framework was developed using the built-in electrolyte Non-Random Two-Liquid (e-NRTL)[39] model in Aspen Plus as a starting point while incorporating a parameter selection methodology based upon the Akaike Information Criterion (AIC).[40] Parameter regression for this framework incorporated data for VLE, heat capacity, and heat of absorption of the ternary system as well as VLE for the binary MEA-H_2O system.

The absorber and stripper are modeled as rate-based packed columns. The absorber is discretized with 30 stages for each bed, and the beds are separated by intercoolers, which are represented by pumparounds with specified solvent flow rate and return temperature. The stripper packing is discretized with 40 stages overall. The number of discretization is determined based on the sensitivity studies. In the absorber column, the speciation reactions are modeled within a liquid film discretized into 20 sections, using the same segmentation given in the work of Plaza.[6] As described in our previous work on thermodynamic modeling of the MEA system,[4] the reaction kinetics are taken from the work of Plaza[6] and modified to the form suggested by Mathias and Gilmartin[41] to ensure thermodynamic consistency when performing uncertainty quantification. Absorber intercooling is modeled using the "Pumparound" option in Aspen Plus, with values of intercooling flow rate and return temperature specified. Heat loss in the stripper column is assumed to be negligible due to the presence of insulation around the column. As described in Part 1 of this work, mass transfer and hydraulic models have been developed for the specific packing used in the absorber and stripper (MellapakPlus 252Y) columns at NCCC, for which models were previously unavailable. FORTRAN user models have been developed and incorporated into the process model for submodels that could not be directly implemented through the in-built options in Aspen Plus. These include liquid property models (viscosity, molar volume, surface tension), interfacial area, reaction kinetics, and pressure drop/holdup.

Development of each of the submodels includes a UQ methodology in which Bayesian inference is used to estimate reasonable distributions for the model parameters. For UQ of the complete capture process, samples from the distributions are drawn and propagated through the process model to obtain estimates of the uncertainty in key process output variables. Our previous work concluded that propagating the parametric uncertainty for viscosity, molar volume, and surface tension models has a negligible effect on the uncertainty of the column model compared to the uncertainty for the thermodynamic framework. For this reason, only thermodynamic model uncertainty is included from the physical property model perspective. The thermodynamic model parameters included in this work, along with their deterministic model values, are given in Table 2 along with their estimated PDFs for the single parameter marginal distributions in Figure 3. The parameter names are assigned based upon those given by Aspen Plus. Further details regarding the e-NRTL model equations and selection of these parameters for inclusion in UQ is given in our work on thermodynamic submodel development.[4]

Table 2. Thermodynamic Model Parameters Considered for UQ

parameter no.	parameter name	value
	electrolyte species formation parameters (MJ/kmol)	
1	DGAQFM (MEA+)	−190
2	DGAQFM (MEACOO−)	−492
3	DHAQFM (MEA+)	−330
4	DHAQFM (MEACOO−)	−691
	Henry's constant parameters (Pa)	
5	HENRY/1 (MEA-H_2O)	28.6
6	HENRY/2 (MEA-H_2O)	−7610
	NRTL molecule−molecule binary parameters	
7	NRTL/1 (MEA-H_2O)	3.25
8	NRTL/1 (H_2O-MEA)	4.34
9	NRTL/2 (H_2O-MEA)	−2200

The PDFs in Figure 3 represent only the single parameter marginal distributions for the thermodynamic model parameters. The overall parameter distribution is a nine-parameter joint distribution, which is not included here as it cannot be visualized. The deterministic values of the parameters for the mass transfer and column hydraulics submodels are given in Table 3.

These parameters and the equations containing them are described in Part 1 of this paper. While propagating the parametric uncertainty through the process model, the values of the mass transfer and hydraulics model parameters are modified simultaneously for the absorber and stripper columns. When evaluating the stochastic model for each case, a sample from the joint distribution of the thermodynamic submodel parameters is drawn and concatenated with a sample of the same size from the joint distributions of the mass transfer parameters and hydraulics submodel parameters. The overall parametric uncertainty is propagated through the process model by evaluating the model for each observation in the sample of the parameter distribution.

2.4. Accounting for Solvent Composition Measurement Error. To effectively compare model predictions with measured data from the process plant, it is important to

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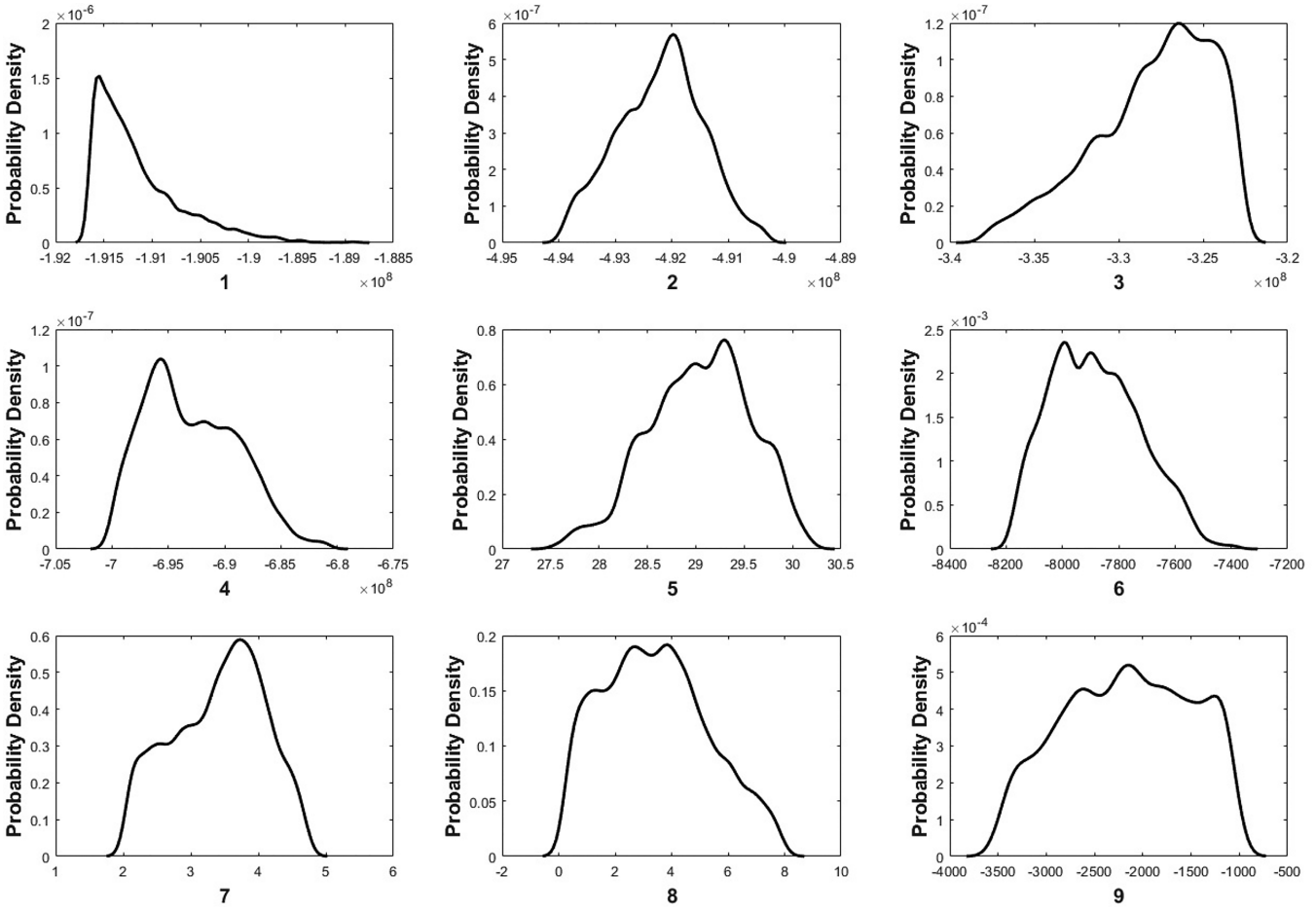


Figure 3. Estimated PDFs for single parameter marginal distributions of thermodynamic model parameters. Parameter numbers correspond to Table 3.

Table 3. Mass Transfer and Hydraulics Model Parameters Considered for UQ

parameter no.	submodel/parameter name	deterministic value
10	interfacial area (A_I)	1.42
11	liquid mass transfer (C_L)	0.203
12	liquid holdup (H_{L1})	11.45
13	liquid holdup (H_{L2})	0.647

estimate the uncertainty of the measured data. Lean and rich solvent amine concentration and CO_2 loading are critical measurements for comparison because of their importance for CO_2 mass balance and absorption/desorption modeling. Therefore, a rigorous measurement uncertainty estimation study was carried out for all amine concentration and CO_2 loading measurements.

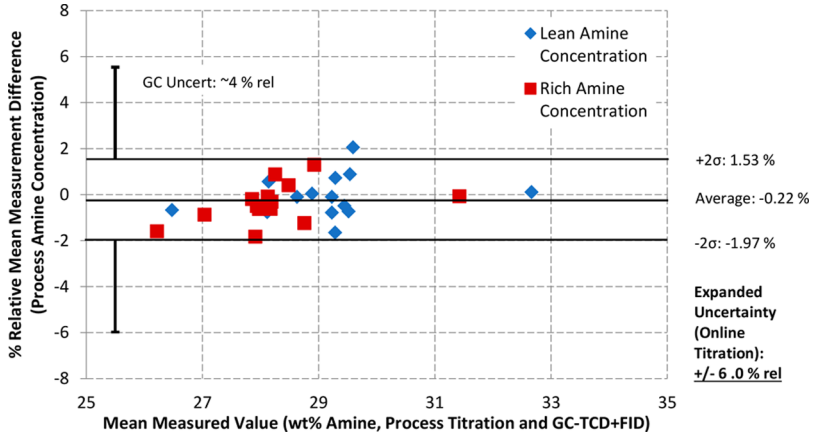


Figure 4. Comparison of process amine concentration measurements with a laboratory gas chromatography method. The horizontal black bars represent the average percent relative mean measurement difference and ±2 standard deviations of the 28 point comparison set. The vertical black bars represent the additional uncertainty added from the gas chromatography comparison method.

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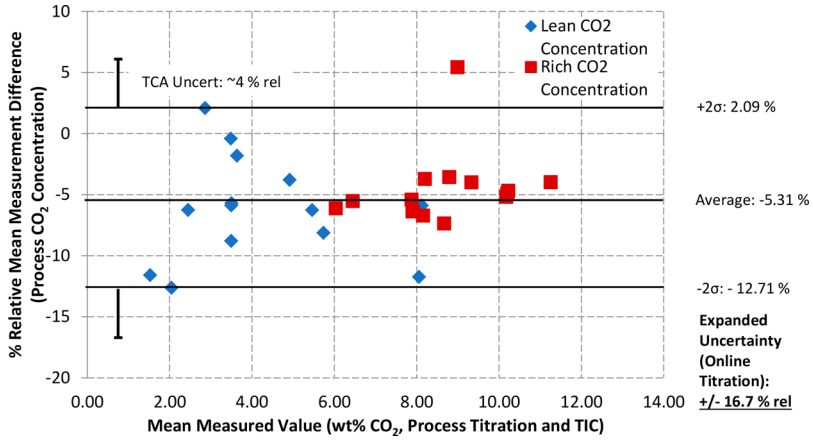


Figure 5. Comparison of process CO_2 concentration measurements with a laboratory gas chromatography method. The horizontal black bars represent the average percent relative mean measurement difference and ±2 standard deviations of the 28 point comparison set. The vertical black bars represent the additional uncertainty added from the gas chromatography comparison method.

Solvent amine concentration was measured periodically (every 70 min) using an automated titration system. Amine concentration was determined using a conductometric equivalence point titration using 0.1 M HCl acid. The method assumes the total alkalinity observed at the equivalence point in the titration approximates the MEA species with a stoichiometry of 1:1 mol MEA/mol acid.

Solvent CO_2 concentration was also measured periodically (every 70 min) using the same automated titration system. The CO_2 concentration was also determined using a conductometric equivalence point titration of a solvent sample dissolved in excess methanol using 0.1 M NaOH base. The method assumes the total acidity observed by equivalence point in the titration approximates the CO_2 species with a stoichiometry of 1:1 mol CO_2/mol base.

The uncertainties of the automated titration of amine concentration and CO_2 concentration methods were estimated by repeatedly comparing with laboratory methods that were considered more accurate and less susceptible to errors when measuring aged solvents. In the case of amine concentration, the reference method was gas chromatography utilizing thermal conductivity and flame ionization detectors (GC-TCD+FID) to quantify the concentrations of amine and water present in the solvent. The uncertainty of this chromatography method was estimated to be ±4% relative to the measured value. In the case of CO_2 concentration, the reference method was a total inorganic carbon (TIC) method that first separates CO_2 from the solvent with the addition of excess strong acid then uses N_2 gas to sweep the CO_2 from the solvent where an IR detector is used for quantification. The uncertainty of this TIC method was also estimated to be ±4% relative to the measured value.

Twenty-eight solvent samples, taken during steady state campaigns, K1−K17, were compared for both amine concentration and CO_2 concentration. The results for these amine concentration and CO_2 concentration comparisons are shown as Bland-Altman plots[42] in Figures 4 and 5.

As seen in Figure 4, the online amine concentration measurements agree with the laboratory method within approximately 2% of the measured value. The online amine concentration measurements appear to regularly read both higher and lower than the laboratory method and the comparison does not appear to be biased by the amine concentration or CO_2 loading of the solvent. Therefore, a reasonable estimate of online amine concentration expanded uncertainty (~95% confidence, k = 2) could be considered approximately 6%.

As seen in Figure 5, the online CO_2 concentration measurement appears to have a bias, regularly reading lower than the laboratory method by an average of approximately 5.3%. Therefore, a reasonable estimate of process CO_2 concentration uncertainty (~95% confidence, k = 2) could be considered approximately 16.7%.

Having estimated the expanded measurement uncertainty for online amine concentration (6% relative) and online CO_2 concentration (16.7% relative), it is possible to derive an expanded uncertainty estimate for nominal amine concentration and CO_2 loading analytically using the relations given in eqs 1 and 2.

Here, w_i and MW_i refer to the weight fraction and molecular weight of species i, respectively. The results of this analysis are given in Table 4.

Table 4. Expanded Uncertainty Estimates for Process Solvent Measurements and Derived Solvent Composition Data during the Steady State Campaign

solvent composition uncertainty estimates—steady state campaign		
solvent measurement	expanded uncertainty process amine concentration	expanded uncertainty process CO_2 concentration
	6.0%	16.7%
derived data	expanded uncertainty nominal amine concentration	expanded uncertainty CO_2 loading
	7.3%	10.7%

The expanded uncertainties in Table 4 have ~95% confidence level (k = 2) and are expressed as a percentage of the measured value.

3. RESULTS AND DISCUSSION

3.1. Deterministic Model. For each case described in Table 1, the process model is evaluated twice. In the first simulation, the flow rates and compositions of the lean solvent and flue gas streams are set based on the information in Table 1. In the second simulation, the estimated composition

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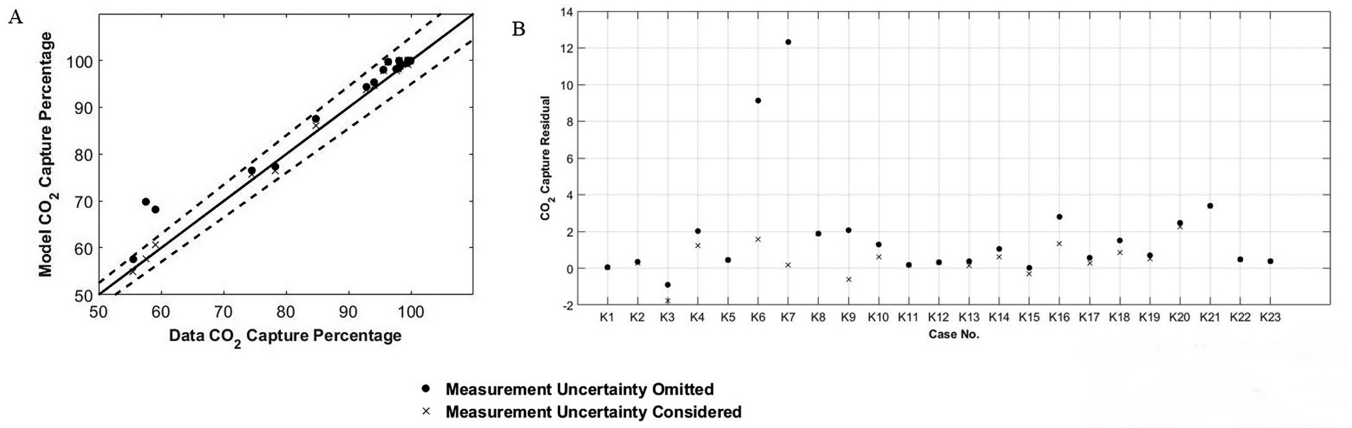


Figure 6. (A) Parity plot and (B) residual plot for gas side CO_2 capture predicted by absorber model.

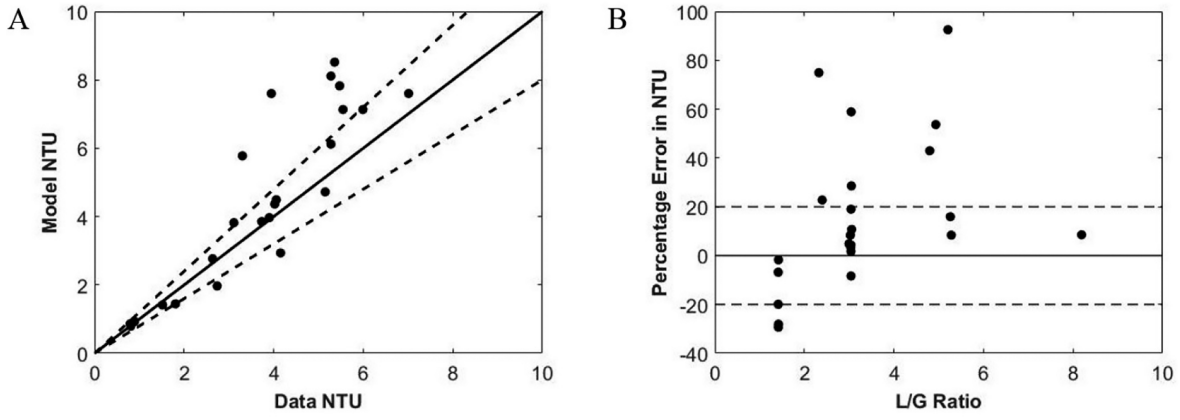


Figure 7. Comparison of data and model NTU in terms of (A) a parity plot and (B) model error as a function of L/G ratio.

measurement discrepancy, as shown in Figures 4 and 5, is used to calculate the input composition of the lean solvent. It is assumed that the actual values of the weight percentage of MEA and the CO_2 are related to the values given in the experimental data by the relationship:

$$\frac{\gamma_i}{100} = \frac{\hat{W}_i - W_i}{\hat{W}_i} \tag{3}$$

where $\hat{W}_i$ is the data value of the weight percentage of species i and the W_i is the hypothetical true value calculated with respect to a measurement error (γ_i) percentage that is treated as a variable in this work. Average values of the measurement error are estimated to be $\gamma^*_{MEA} = -0.22$ and $\gamma^*_{CO_2} = -5.31$ from the Bland-Altman plots in Figures 4 and 5. The model is simulated for each case with the baseline lean solvent composition as well as the modified composition (calculated from the baseline values of γ_i^* for MEA and CO_2). The simulation is performed for cases with ($\gamma_i = \gamma_i^*$) and without ($\gamma_i = 0$) inclusion of composition measurement uncertainty in the simulation input. In both cases, the flow rate and composition for the rich solvent into the stripper column are inherited from the results of the absorber simulation. Since the objective of this work is to validate the process model with the data from NCCC in terms of the absorber and stripper performance, some simplifications have been made for modeling the overall system. A rigorous model of the lean-rich heat exchanger is not included in this work, and the rich solvent temperature in the stripper inlet is specified based on the plant data. Also, the recirculation of the lean solvent from the stripper outlet to the absorber inlet has not been modeled here. Moreover, these simplifications are useful in reducing the computational time of the stochastic model evaluation, which will be discussed in the next section.

A parity plot (with dashed lines representing 5% error) and residual plot of the model and data values of CO_2 capture is shown in Figure 6 for the simulations with and without consideration of composition uncertainty. The residual is represented as $y_{\text{model}} - y_{\text{data}}$, a convention followed throughout this work for calculations of residuals and error percentages. The data values of CO_2 capture shown here are calculated from the gas-side measurements, which are generally more consistent with the model predictions in comparison with the liquid-side measurements. A table showing the comparison of the liquid and gas-side CO_2 capture values to the model predictions is given in Table S3. No model parameters are manipulated to improve the fit of the process model to the pilot plant data since it was desired to evaluate the predictive capability of the model.

When no measurement discrepancy is taken into consideration, the sum of square error (SSE) between model predictions and plant data of CO_2 capture percentage is 280.2. This value decreases to 32.0 when composition measurement discrepancy is included in the simulation. From Figure 6B, it is shown that the CO_2 capture percentage residuals are generally larger when composition discrepancy is not considered. The

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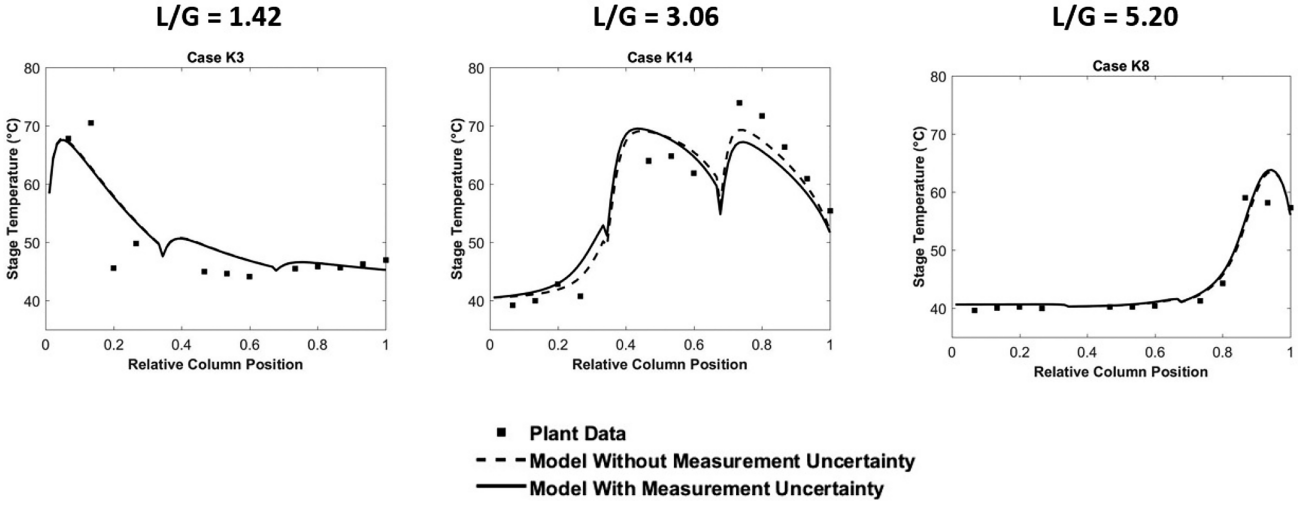


Figure 8. Comparison of model and data temperature profiles for absorber operation with three beds and intercooling.

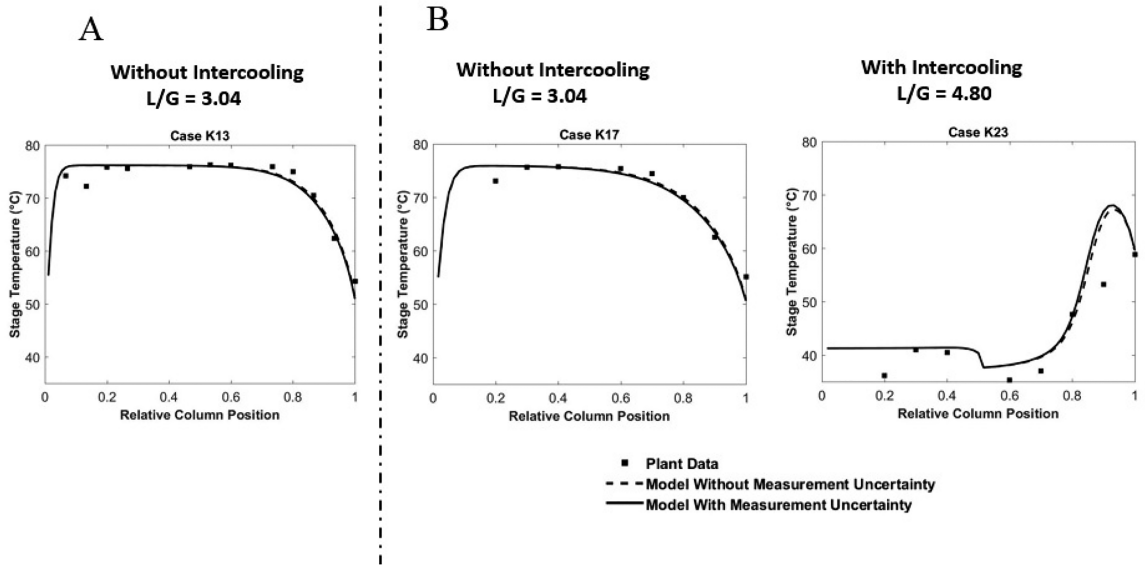


Figure 9. Comparison of model and data temperature profiles for absorber operation with (A) three and (B) two beds.

overall decrease in the error of the CO_2 capture percentage when considering measurement uncertainty is not statistically significant for the set of 23 cases, as a paired *t* test between the absolute value of the residual error for the two cases reports $p = 0.085$. Most of the improvement in the model fit that occurs when taking concentration measurement uncertainty into consideration may be attributed to cases in which the CO_2 loading in the inlet solvent to the absorber is relatively high (i.e., cases K6 and K7). Since the discrepancy in solvent CO_2 concentration is estimated to be a fixed percentage, its effect is naturally greater on cases with higher CO_2 concentration in the inlet solvent.

The CO_2 removal in the absorber may also be represented in terms of the theoretical number of transfer units (NTU) for the separation, and this convention is preferred in some of the existing literature on the CO_2 capture process.[43] This metric is related to the CO_2 capture percentage by

$$\mathrm{NTU} = -\ln\left(1 - \frac{\%\mathrm{CO_2\ capture}}{100}\right) \quad (4)$$

The NTU comparison is shown in Figure 7; for simplification, the model predictions are only shown for the case in which measurement uncertainty in the solvent composition is considered. In addition to the parity plot, the percentage error in the NTU predictions is shown in relation to the liquid to gas mass ratio for each absorber case.

In Figure 7A, the dashed lines represent ±20% error, and the majority of the model predictions lie within this range. In Figure 7B, the percentage error in NTU is shown to vary widely over the range of L/G ratio, although it is generally underpredicted at low L/G (<2) and overpredicted elsewhere. Increasing L/G is associated with increasing NTU, although the influence of other important absorber variables (lean loading, CO_2 concentration in flue gas) is not shown in these figures. It should be noted that consideration of NTU when analyzing model performance results in a heavier penalty placed on deviance between model and data at larger values of CO_2 capture percentage/NTU. For the data shown in this work, most of the test conditions for which the discrepancy between data and model is largest occur at NTU greater than 3.5 (97% CO_2 capture). Due to very high sensitivity of NTU to

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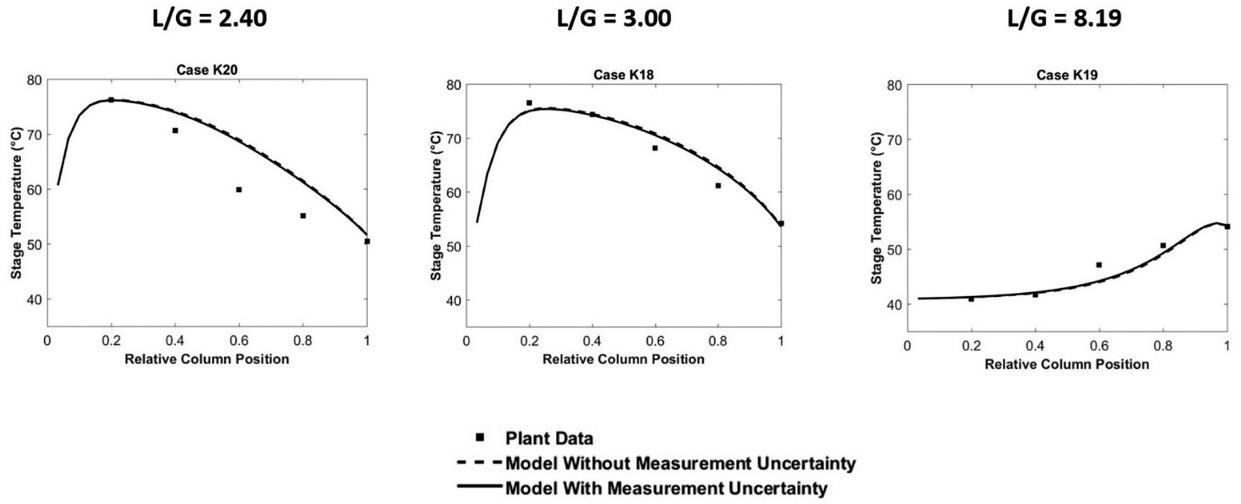


Figure 10. Comparison of model and data temperature profiles for absorber operation with single bed.

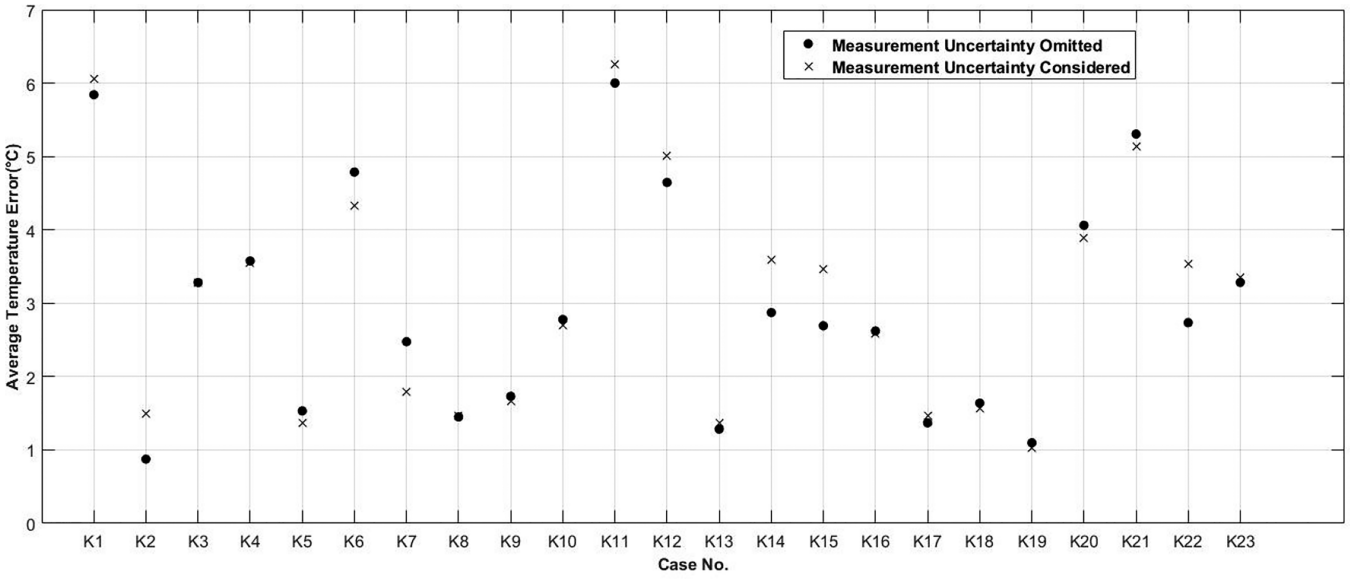


Figure 11. Values of temperature profile error (eq 5) calculated for absorber column simulations with and without consideration of measurement error uncertainty.

CO_2 capture percentage in this regime, it is increasingly difficult to distinguish whether discrepancy in experimental and model NTU may be attributed to inaccuracy in the model or the measurement technique. Accordingly, such data may not be particularly useful for model validation purposes unless the measurement technique(s) for determining CO_2 capture at such deep removal is highly accurate and reproducible. Furthermore, such deep removal of CO_2 is of little practical interest for postcombustion CO_2 capture. Since pilot plant campaigns are costly and time-consuming, efforts will be made in the future to avoid a large amount of data collection for such deep removal.

Temperature profile data were also provided for the absorber and stripper columns during the test campaign at NCCC. For select cases, the comparison of the model and experimental absorber temperature profiles is given in Figures 8, 9, and 10 for the model with and without the assumption of solvent concentration discrepancy. In all figures, the relative column positions of 0 and 1 represent the top and bottom of the column, respectively. The comparisons for all 23 cases are available in Figures S1−S4.

As shown in the preceding figures, the temperature profile in the absorber may assume various shapes, depending on the L:G ratio, the inlet solvent loading, and the number of beds and intercoolers. The temperature bulges in the column, which occur as a result of the exothermic reaction between CO_2 and MEA, are generally located toward the top of the column during operation at low L:G ratios and toward the bottom for high L:G ratios. For most cases, the model is capable of accurately predicting the shapes of the various temperature profiles, and the consideration of concentration measurement discrepancy does not appear to have a large impact on the predicted temperature profile. To quantify the quality of the model fit of the temperature profile, an average temperature profile error is evaluated for each case, defined as

$$\hat{T}_{\text{error}} = \frac{\sum_{i=1}^{n} |T_i - \overline{T}_i|}{n} \quad (5)$$

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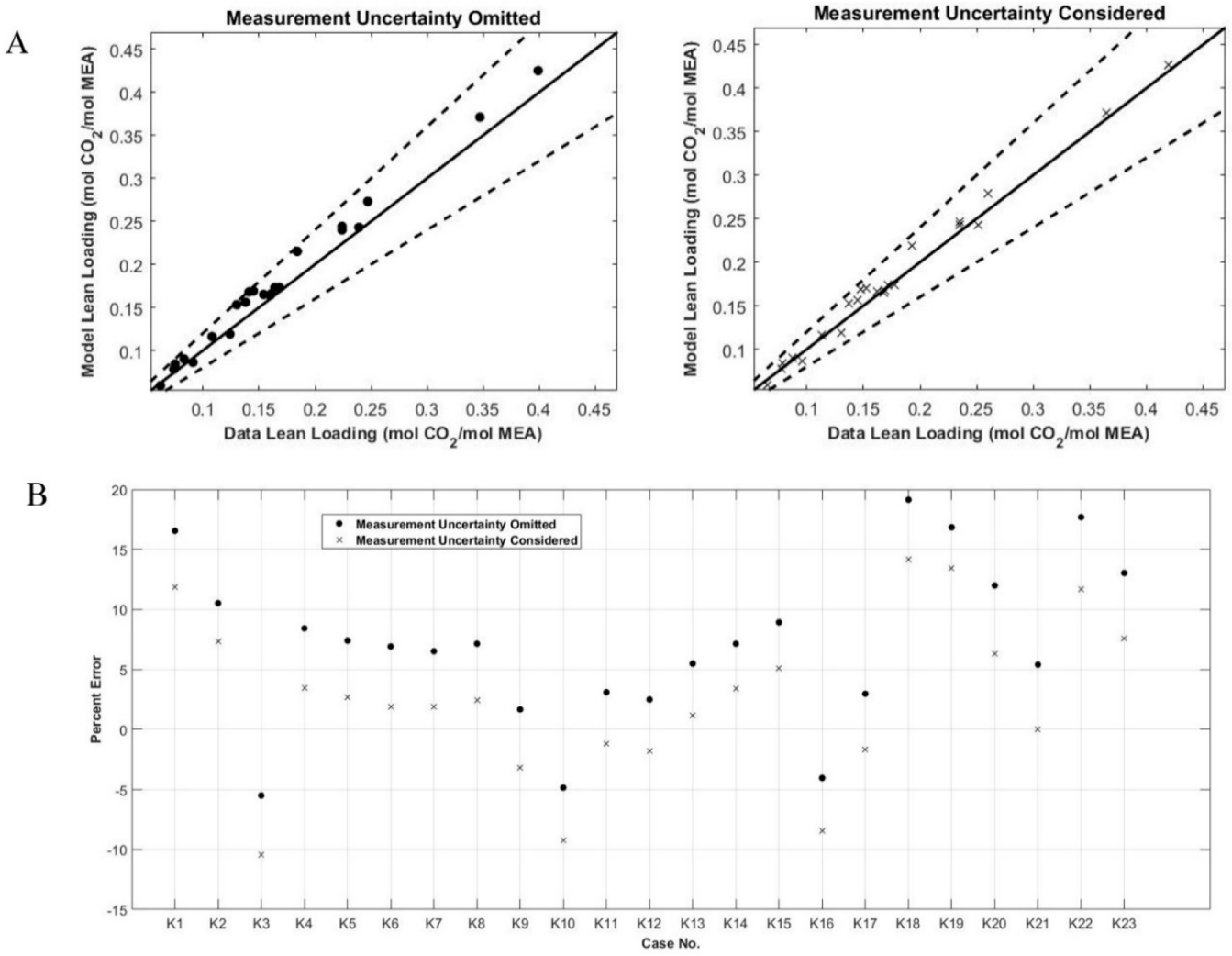


Figure 12. Comparison of experimental data and model prediction of CO_2 loading in the regenerator solvent outlet.

where T_i and $\overline{T}_i$ are the model and data values, respectively, of temperature at a given point in the column for which a measurement is given, and n is the total number of such points. The value of the absorber simulation for each case is shown in Figure 11.

For the set of 23 cases, the average temperature error in the absorber column is 2.822 ± 1.520 °C for the process model in which measurement uncertainty is not considered. This increases to 2.973 ± 1.513 °C when measurement uncertainty is considered. A paired t test finds no significant difference between the average temperature error in the predictions of the cases for model evaluation (p = 0.1773). For some of the cases for which this value is higher (e.g., K1, K11, and K12), the error may be attributed to a discrepancy in the predicted location of the temperature bulge in the middle of the column, while the match of the temperature profile near the top and bottom of the column is quite accurate. Possible sources of this discrepancy include uncertainty in plant measurements that are used as specifications for model boundary conditions as well as simplifications in the model itself. For example, the column packing is modeled as a single section without inclusion of rigorous distribution and redistribution submodels. Discussion of a further source of uncertainty, the submodel parameters, and its effect on predictions of plant variables, including temperature profiles, is forthcoming.

The comparison of the experimental values and model predictions of the CO_2 loading of the lean solvent from the stripper outlet is given in Figure 12; the dashed lines in the parity plot represent ±20% error. Since the lean loading in the stripper outlet depends on the assumed value of measurement uncertainty, the experimental data values as well as the model values are different for the two cases (with and without composition discrepancy). The experimental values of the lean loading as well as the model values for both cases are tabulated in Table S4.

The fit of the model and experimental lean loading is shown to be within ±10% error for the majority of the test runs, and the absolute value of the percent error does not exceed 20% for any of the test runs. For the case in which solvent composition discrepancy is neglected, the SSE of the lean loading prediction is 0.314, and this value decreases to 0.206 upon inclusion of the discrepancy. The percentage error for the entire set of data points is compared for the two cases using a paired t test, which shows that the error significantly decreases when measurement discrepancy is incorporated into the model (p = 0.0008). The comparison of the model and data temperature profiles is shown in Figure 13. Selected temperature profiles

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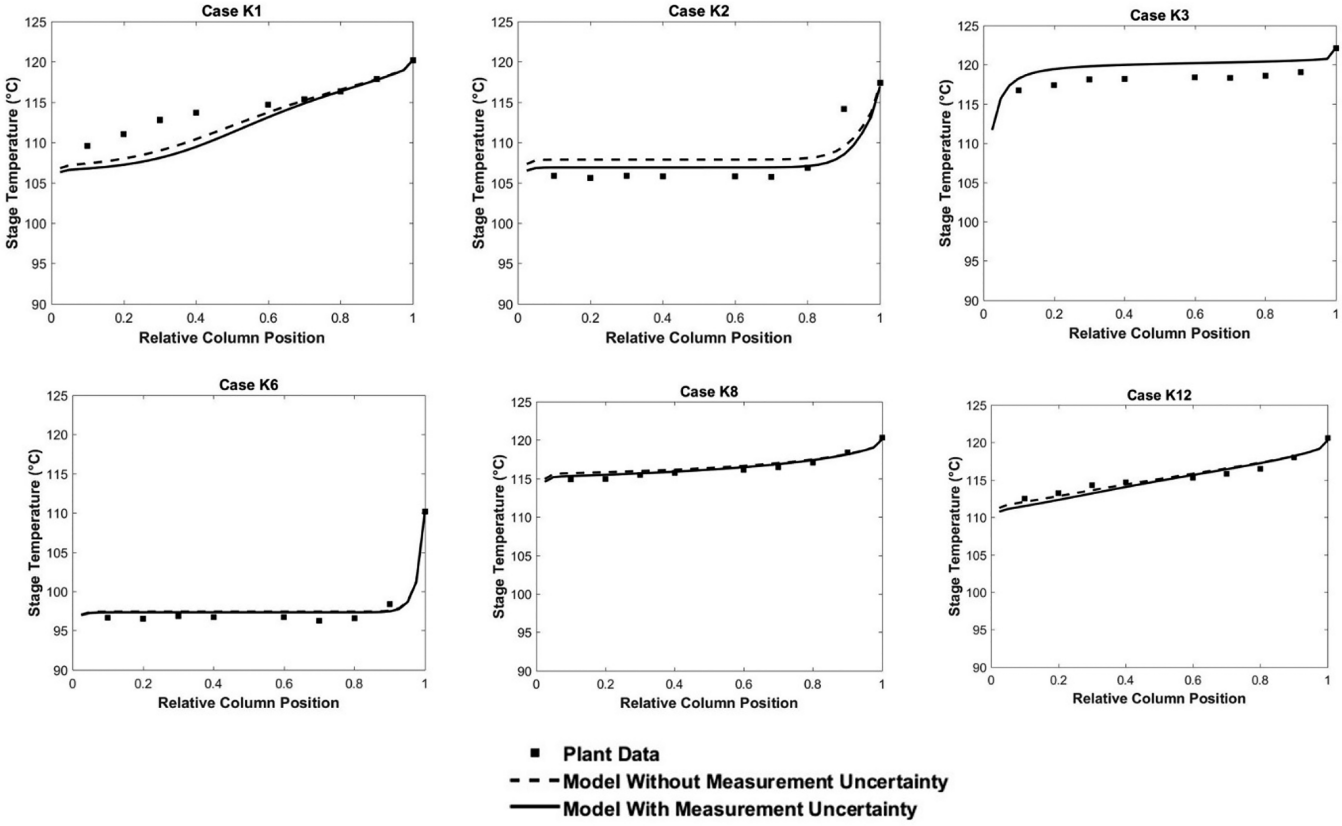


Figure 13. Comparison of model and data temperature profiles for stripper for selected cases.

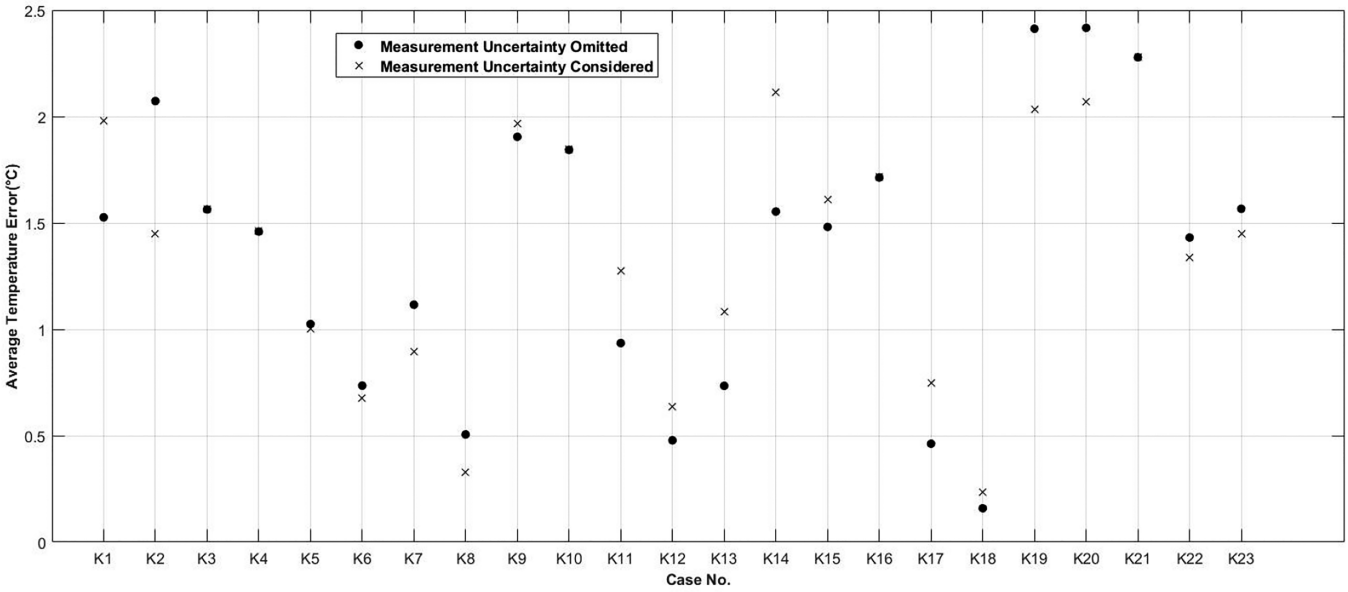


Figure 14. Values of temperature profile error (eq 5) calculated for stripper column simulation. Red and blue squares represent simulations with and without consideration of composition measurement uncertainty.

are shown here, and the complete set is presented in the Figures S5–S8.

The shape of the temperature profile for the stripper is also dependent on the operating variables, notably the rich solvent flow rate, CO_2 loading, and the reboiler duty. The simulated temperature profiles match the experimental profiles accurately for most cases, although the column temperature is overestimated for many cases to a small extent (e.g., K3 as shown here). The temperature of the solvent exiting the column, given as the temperature value at a relative column position of 1, is captured accurately by the model for nearly all cases, although less so for the later cases (K19–K23, shown only in the Supporting Information). Note that cases K18–K23 were conducted separately, following the dynamic test runs, and there could be issues due to solvent degradation that were not present for the first set of cases where the fresh solvent was

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Table 5. Values of Sobol Indices for Parameters with Respect to CO_2 Capture Percentage Prediction for Three Cases

parameter number	parameter name	K1	K6	K18
		thermodynamic model		
1	DGAQFM (MEA+)	0.473	0.0544	5.76×10^{-3}
2	DGAQFM (MEACOO−)	0.0445	0.0183	6.82×10^{-3}
3	DHAQFM (MEA+)	9.04×10^{-4}	0.0138	3.89×10^{-3}
4	DHAQFM (MEACOO−)	0.0221	2.00×10^{-4}	4.43×10^{-3}
5	HENRY/1 (MEA-H_2O)	0.114	0.348	0.108
6	HENRY/2 (MEA-H_2O)	0.0945	0.275	0.0747
7	NRTL/1 (MEA-H_2O)	1.27×10^{-4}	3.71×10^{-3}	0.0483
8	NRTL/1 (H_2O-MEA)	0.699	0.221	0.560
9	NRTL/2 (H_2O-MEA)	0.578	0.228	0.580
		integrated mass transfer model		
10	A_I (interfacial area)	0.0179	8.38×10^{-3}	0.0271
11	C_L (liquid mass transfer)	0.0274	3.12×10^{-3}	8.24×10^{-3}
		holdup model		
12	H_{L1} (liquid holdup)	2.41×10^{-4}	1.35×10^{-3}	4.50×10^{-4}
13	H_{L2} (liquid holdup)	5.27×10^{-3}	3.19×10^{-3}	1.00×10^{-5}

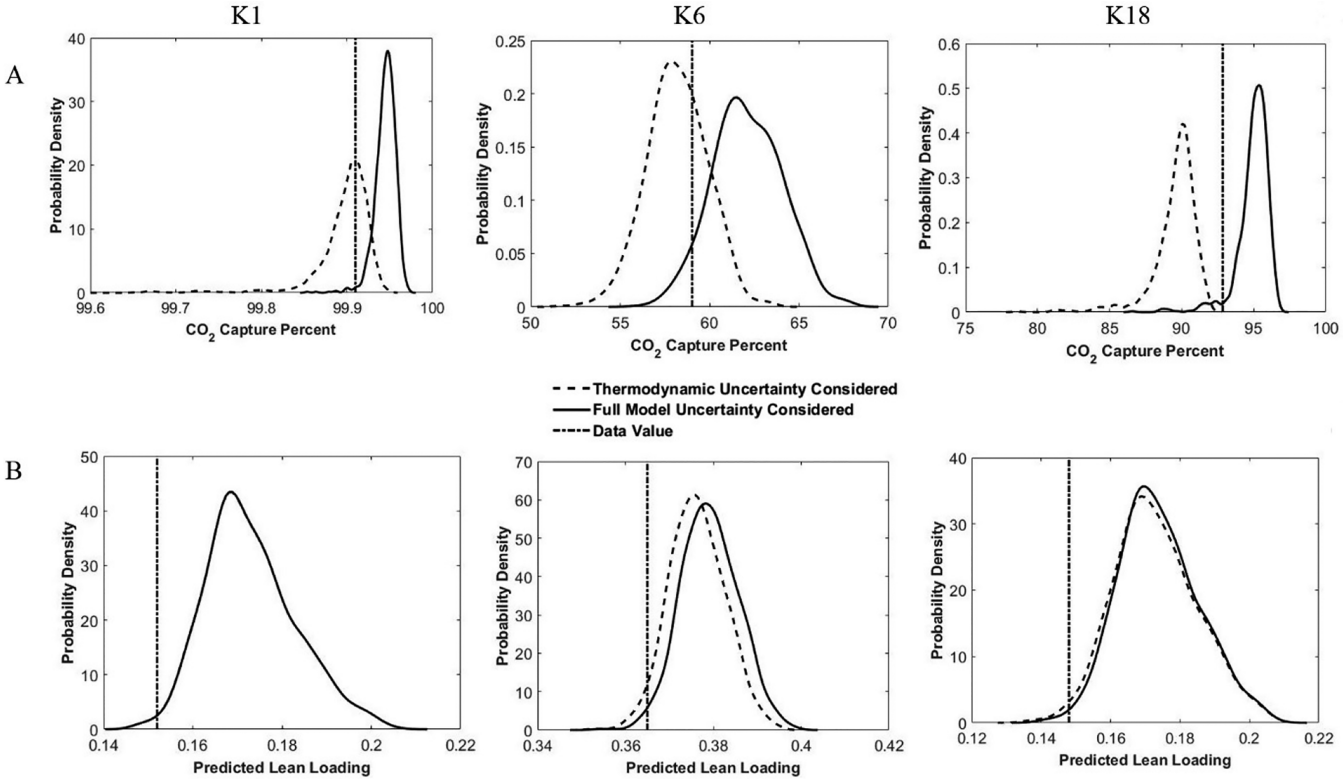


Figure 15. Estimated PDFs resulting from propagating parametric uncertainty through process model for three cases. PDFs for CO_2 capture efficiency are given in (A) and predicted lean loading in (B), and data values are shown for comparison.

used. Future work will address the effect of degradation products on the performance of the solvent. The temperature profile error is calculated for the stripper column simulations and shown in Figure 14.

The average temperature profile error is similar for the cases in which composition measurement uncertainty is neglected (1.364 ± 0.650) and included (1.381 ± 0.593).

3.2. Stochastic Model. While the previous results are focused on deterministic model comparison with the plant data from NCCC, the inclusion of UQ in the submodels allows for probabilistic comparison of this data with the model. Essentially, the model is evaluated stochastically, so that ranges of important process outputs (CO_2 capture percentage, lean loading in stripper outlet) may be predicted and used to validate the process model. The stochastic model is evaluated with a Monte Carlo approach, in which the process model is simulated a large number (N) of times, with varying submodel parameters. For each simulation, the values of the submodel parameters are drawn from the parameter distributions, which have been discussed previously in section 2.3 of this paper. The result is a set of values of the output variables that may be used to estimate probability density functions (PDFs), confidence intervals, and other related statistics.

When evaluating the stochastic model for absorber and stripper column simulations, a sample size of $N = 1000$ drawn from the parameter distributions is found to be adequate for

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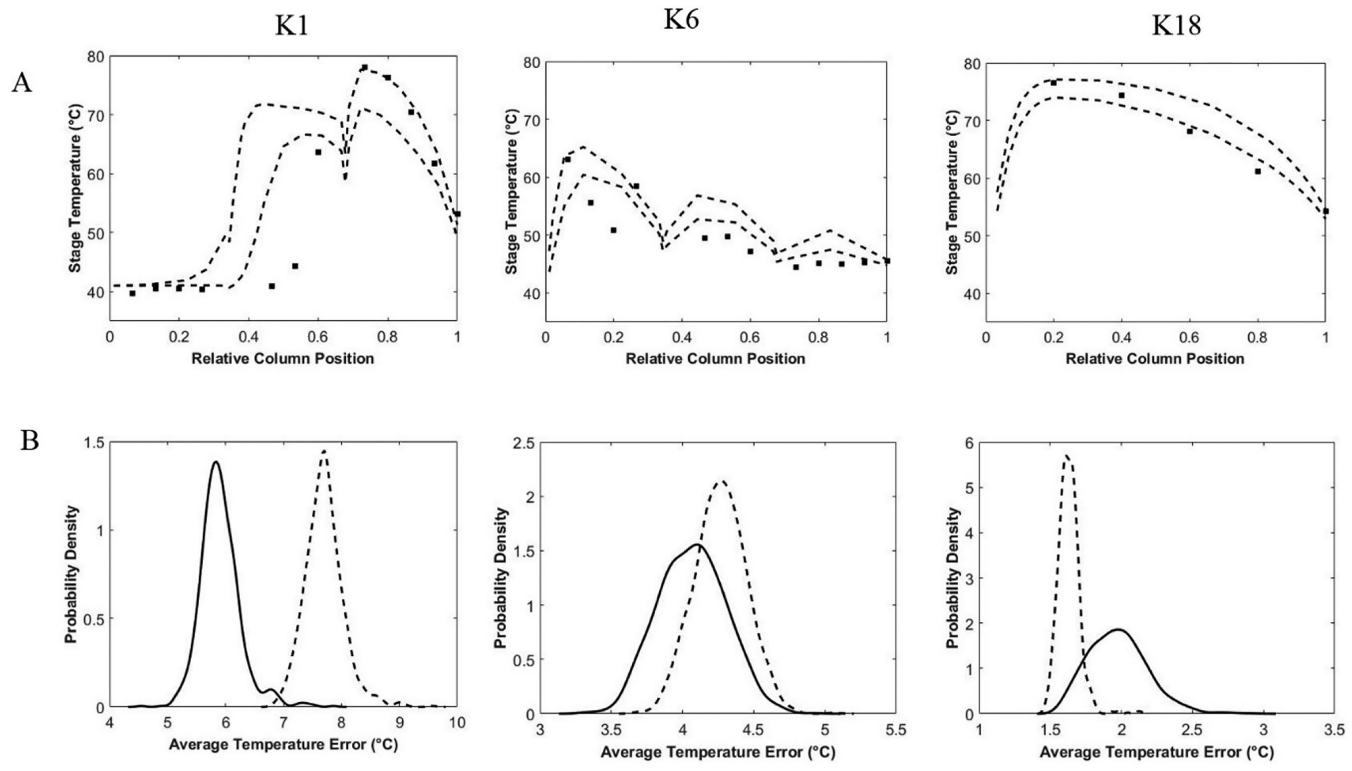


Figure 16. Calculated temperature profiles for absorber column stochastic model. (A) Temperature profile data compared to upper and lower limits of stochastic model temperature predictions. (B) Estimated PDFs for average temperature profile error for stochastic simulation. Dashed lines represent cases in which thermodynamic model parameter uncertainty is considered and solid lines represent cases in which thermodynamic and mass transfer model uncertainty are considered.

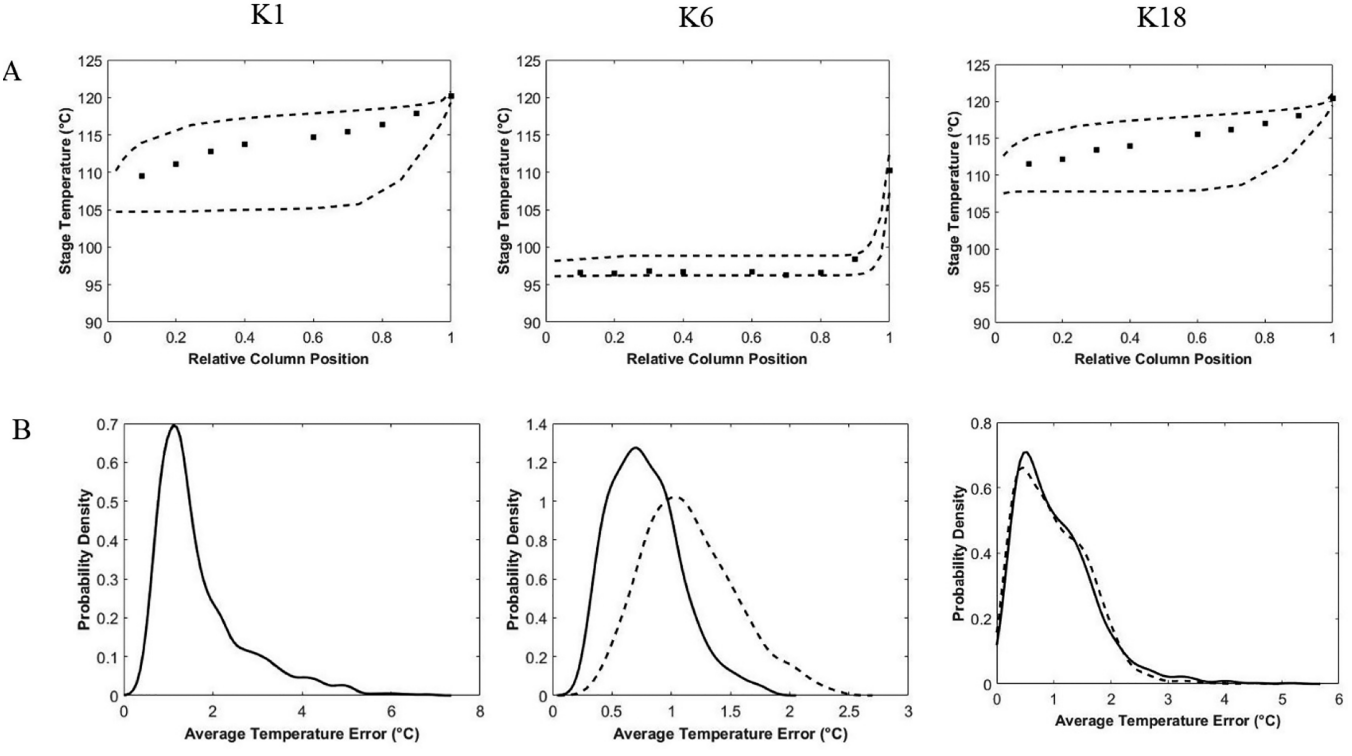


Figure 17. Calculated temperature profiles for stripper column stochastic model. (A) Temperature profile data compared to upper and lower limits of stochastic model temperature predictions. (B) Estimated PDFs for average temperature profile error for stochastic simulation. Dashed lines represent cases in which thermodynamic model parameter uncertainty is considered and solid lines represent cases in which thermodynamic and mass transfer model uncertainty are considered.

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Table 6. Values of Sobol Indices for Parameters with Respect to Lean Loading Prediction for Three Cases

parameter number	parameter name	K1	K6	K18
		thermodynamic model		
1	DGAQFM (MEA+)	0.0278	0.0357	0.0272
2	DGAQFM (MEACOO−)	0.0300	0.0168	0.0328
3	DHAQFM (MEA+)	0.0377	0.0297	0.0312
4	DHAQFM (MEACOO−)	0.0344	0.0145	0.0360
5	HENRY/1 (MEA-H_2O)	0.510	0.464	0.517
6	HENRY/2 (MEA-H_2O)	0.330	0.321	0.329
7	NRTL/1 (MEA-H_2O)	4.90×10^{-4}	1.55×10^{-3}	3.40×10^{-4}
8	NRTL/1 (H_2O-MEA)	0.0495	0.106	0.0464
9	NRTL/2 (H_2O-MEA)	0.0509	0.0983	0.0386
		integrated mass transfer model		
10	A_I (interfacial area)	0	4.30×10^{-5}	0
11	C_L (liquid mass transfer)	0	5.40×10^{-5}	0
		holdup model		
12	H_{L1} (liquid holdup)	0	0	0
13	H_{L2} (liquid holdup)	0	3.90×10^{-5}	0

characterizing the resulting distributions of CO_2 capture percentage and lean loading. For the purpose of brevity, results are shown only for three of the data cases (K1, K6, and K18). In this portion of the work, composition measurement uncertainty is included in all simulations. Two parameter distributions are propagated through the model: the thermodynamic parameter set (the parameters listed in Table 4) and the full parameter set (all parameters listed in Tables 4 and 5). In the former case, mass transfer and hydraulic model parameters are represented only by their deterministic values. In each data case, 1000 simulations are performed, with variation in the corresponding parameter values of interest. The PDFs for CO_2 capture efficiency and lean loading in the stripper outlet are estimated and presented in Figure 15.

The results given in Figure 15 allow for comparison of the effect of thermodynamic and mass transfer/hydraulics model uncertainty separately. For all three cases, the modal value of the CO_2 capture efficiency is shifted upward as a result of the inclusion of the mass transfer and hydraulics submodel uncertainty. As shown in Part 1 of this work, the majority of the probability density of the interfacial area parameter A_I is shifted to a higher value as a result of the Bayesian inference, with most of the probability density concentrated at values higher than the deterministic value of this parameter. Consequentially, an increase in the CO_2 capture occurs along with the increase in the interfacial area parameter.

As expected, the performance of the stripper, which operates close to equilibrium, remains relatively unaffected by mass transfer model uncertainty. This is evident by the minimal change in the distributions of outlet lean loading for the three cases. The slightly perceptible change in the distribution estimated for case K6, and case K18 to a lesser extent, which is shown in Figure 15, may be attributed to the change in the distributions of rich solvent flow rate and CO_2 loading that result from the absorber model simulation. As the absorber model operates near complete CO_2 capture for case K1 regardless of parametric uncertainty, the variation in rich solvent flow rate and composition is not large enough to have an appreciable effect on the stripper simulation. For all cases shown, the data values for CO_2 capture percentage and lean loading are included within the ranges predicted by the stochastic model. For the lean loading, however, the data values lie toward the lower limit of the stochastic model prediction. This is consistent with the model's slight overprediction of this variable, which was previously shown in Figure 12.

The temperature profiles for absorber and stripper are also evaluated for the stochastic model, and the results are shown in Figures 16 and 17, respectively.

The temperature profile estimations for both columns tend to take on a rather wide range in many of the cases. For the stripper, a wide range for the temperature within the column is shown, although the temperature of the solvent outlet at the bottom of the column has a very narrow range in the stochastic simulation and matches the data well for these cases. As for the process variables which are shown in Figure 15, the distributions of the average temperature profile error and the effect of the thermodynamic and mass transfer parametric uncertainty on them are highly case dependent.

3.3. Variance-Based Sensitivity Analysis. In the previous section, parameter distributions from various submodels are propagated through the process model, for different experimental conditions, in order to quantify the effect of parametric uncertainty on the estimated uncertainty of process outputs. In addition to estimating the overall output uncertainty, quantification of the effect of individual sources on this total uncertainty is also important. For instance, certain parameters that have been characterized by uncertainty distributions in the submodel development process may actually have little effect on the process outputs. If such parameters are identified, down-selection of the uncertain parameter space in the model may be possible, and this could result in reduction of the computational expense associated with stochastic evaluation of the process simulation model. Furthermore, knowledge of the relative importance of different sources of uncertainty is essential for the modeling of novel solvent systems, as it can be leveraged in further data collection needed to improve the component submodels.

The effect of various submodels on the overall model uncertainty has already been investigated in an ad hoc manner in the previous section, by estimating the output distributions first with only the thermodynamic model parameters and then demonstrating the effect of incorporating the parameters of the mass transfer and hydraulics submodels. A more quantitative approach is used here, in which the relative contribution of the uncertainty of each parameter to the uncertainty of the process

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variables is estimated through a Sobol sensitivity analysis.[38] In this analysis, a dependent variable that is calculated as a function of k independent variables $[y = f(x_1,x_2,\cdots,x_k)]$ is considered. Sensitivity indices are calculated for the contribution of individual parameters (S_1) and interactions among pairs (S_{ij}) or groups ($S_{1,2,\cdots,k}$) to the overall variability of the dependent variables. These terms satisfy the equation:

$$\sum_{i=1}^{k} S_i + \sum_{1\le i<j\le k} S_{ij} + \ldots + S_{1,2,\ldots,k} = 1 \tag{4a}$$

Since eq 4 includes $2^k - 1$ terms (8191 for this example), total Sobol indices (S_{Ti}) are presented in this work in lieu of the individual values. Each term S_{Ti} represents the total sensitivity of an individual parameter and its aggregate interactions of all orders. These indices are calculated for each of the three cases considered and with respect to the CO_2 capture efficiency and lean loading in the regenerator outlet as separate dependent variables. The output variables are represented as a function of the 13 uncertain parameters for each case through response surface models built with multivariate adaptive regression splines (MARS).[44] The response surface methodology reduces the computational expense of this process by providing a surrogate model for calculating the process outputs for parameter realizations, reducing the required amount of process simulations. The index values are given in Tables 5 and 6 for the CO_2 capture percentage and outlet lean loading from the stripper.

In the absorber simulation, the NRTL parameters (8−9) have the greatest effect on the uncertainty in the CO_2 capture percentage prediction, and the effect of the mass transfer/hydraulics model parameters (10−13) is relatively small for all cases. This is especially true for case K6, in which the CO_2 capture percentage is relatively low and the effect of the mass transfer is diminished. Although the change in the interfacial area parameter occurring from Bayesian inference was shown to have a major effect on the absorber model performance, as shown in Figure 15, the effect of the posterior uncertainty on the absorber model uncertainty is rather small. For the stripper, neither the mass transfer nor the hydraulics model parameters have any appreciable effect on the performance of the column as evident from the negligibly small values of the Sobol indices. The Henry model parameters (5−6) are shown to have the largest effect on the calculation of the lean solvent loading for all cases. The calculation of small Sobol indices for mass transfer and hydraulics model parameters is generally consistent with the results of Figure 15C, in which the PDFs of outlet lean loading change negligibly with the inclusion of mass transfer and hydraulic model uncertainty in the stochastic model.

The parameters for the liquid holdup model are shown to have very low Sobol indices for the absorber and stripper column output variables for the representative cases analyzed in this work. This suggests that it is reasonable to neglect the uncertainty of these model parameters when evaluating the stochastic model, essentially replacing the probability distributions with point values. For a novel solvent system with limited property and process data available, there would likely be a larger set of uncertain parameters determined during the submodel development. In such a case, the Sobol analysis would be extremely valuable for down-selection of the uncertain parameter space for improving computational efficiency of evaluating the stochastic model process simulation. Furthermore, characterization of the major sources of uncertainty may provide insight into which submodels can be refined and improved through additional data collection, thus allowing for prioritization of resources during the development of a novel solvent system. Although the Sobol method is useful for analyzing the relative effects of different submodels on the overall process model uncertainty, one must be careful about making definitive conclusions regarding the relative importance of individual parameters based on the Sobol indices. This aspect needs to be considered, especially where surrogate models are used while estimating the parameters using the Bayesian framework in which parameter distributions are estimated and the Sobol indices are calculated.

4. CONCLUSIONS

A rigorous modeling framework for solvent-based CO_2 capture systems has been proposed in this work and applied to the development of a model for an aqueous MEA. The complete process model of the CO_2 capture process with aqueous MEA solution has been developed by combining physical property, reaction kinetics, mass transfer, and hydraulic submodels developed previously. The overall model has been validated with steady-state data obtained from a MEA solvent test campaign at the National Carbon Capture Center, which was rigorously planned to cover a large range of operating conditions for both the absorber and stripper columns. Through uncertainty quantification, the validation procedure used in this work expands upon the typical approach of comparing plant data to a deterministic process model and verifying the predictivity of the model. By propagating distributions of submodel parameters through the process model, the model uncertainty is taken under consideration and corresponding ranges are predicted for the model outputs. When taking the parametric uncertainty into consideration, the model has been shown to accurately predict large-scale pilot plant data, including absorber CO_2 capture percentage and CO_2 loading in the regenerator outlet as well as the temperature profiles for both columns.

Along with measurement of key process variables, a methodology was used for estimating the uncertainty in solvent composition by comparing the NCCC process measurements of MEA and CO_2 concentration to laboratory methods of a known uncertainty. The consideration of this measurement sensitivity provides insight into the discrepancy between the pilot plant data and the corresponding model predictions. The improvement in the model predictions upon including this analysis has demonstrated the importance of accounting for measurement error and bias when performing model validation.

A variance-based sensitivity analysis, using the method of Sobol, has been utilized in this work to study the effect of the parameters of the different submodels. This analysis has shown that the relative contributions to the key output variable uncertainty of different parameters is highly dependent on the specific output variable of interest as well as the operating regime of the process. This demonstrates the importance of characterizing the uncertainty in all underlying submodels when modeling a solvent-based CO_2 capture process, as rigorous study over many different operating conditions and for various outputs is required for examining the importance of the submodel parameters. Although relatively few model parameters were found to have significant uncertainty for

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this baseline MEA system, models of novel solvent systems, for which submodels will be developed with fewer data, will likely have more uncertain parameters. Accordingly, the calculation of the Sobol indices would be extremely valuable for down-selection of the parameter space for such a model.

In summary, a steady-state model of the MEA-based CO_2 capture process has been developed and validated with large-scale pilot plant data from NCCC. The model has been developed, with uncertainty quantification, using a rigorous framework that can be readily applied to any solvent system for which bench-scale data are available for developing the submodels that comprise the overall model. The MEA model has been validated at the pilot scale, demonstrating the value of this modeling framework. While the model has been shown to be reasonably predictive of the 0.5 MWe NCCC pilot plant over a large range of operating conditions, its performance needs to be investigated at a larger scale for a more complete validation. Future work related to the solvent modeling framework will focus on applying this methodology to novel solvent systems for CO_2 capture applications. In forthcoming work, the stochastic modeling methodology will be expanded to develop a sequential design of experiments methodology (sDoE) that will leverage the knowledge of model parametric uncertainty to plan for additional data collection, which in turn will be used to update the parameter distributions. This work will also initially focus on the baseline MEA system and later be applied to novel solvent systems. Finally, additional work will focus on the development of a dynamic model of the MEA system using the steady-state model developed in this work as a starting point. Lastly, the model described here is available for free download under the "Process Models Bundle" of CCSI toolset available at: https://github.com/CCSI-Toolset/MEA_ssm.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.8b01472.

Some information regarding the data and model comparison for the MEA test campaign at NCCC has been omitted for the purpose of brevity; summary of NCCC steady state data; comparison of model predictions of CO_2 capture efficiency; comparison of model predictions of CO_2 loading; and comparison of model and data absorber temperature profiles (PDF)

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Notes

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